Analysis of a multiscale discontinuous Galerkin method for convection-diffusion problems: a Lean blueprint

Formalization of A. Buffa, T. J. R. Hughes and G. Sangalli (2006)

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Chapter 1

Overview

This blueprint describes a complete Lean 4 formalization of the main results of

> A. Buffa, T. J. R. Hughes, G. Sangalli, *Analysis of a multiscale discontinuous Galerkin method for convection-diffusion problems*, SIAM J. Numer. Anal. **44** (2006), no. 4, 1420–1440.

The paper studies the convection-diffusion problem

$$-\kappa\Delta\varphi + a\cdot\nabla\varphi = f \text{ in } \Omega, \qquad \varphi = g \text{ on } \partial\Omega,$$

discretized by an interior-penalty discontinuous Galerkin (DG) method with upwind fluxes, its streamline-diffusion stabilization (SDG), and the multiscale variant (MDG/SMDG). In the multiscale variant, the discontinuous trial and test functions are obtained from a continuous coarse field by solving local DG problems on the elements. The main results are coercivity and continuity of the DG and SDG forms (Propositions 3.2, 3.3), an inf-sup condition for the DG form in the stronger SDG norm (Theorem 3.5), and stability of the local problems (Lemmas 4.4, 4.5). They also include optimal a-priori error estimates

$$\|\varphi-\varphi_h\|_{\mathrm{SDG}} \lesssim \Big(\sum_{T\in\mathcal{T}_h} \big(|a|\,h_T^{2k+1} + \kappa\,h_T^{2k}\big)|\varphi|^2_{H^{k+1}(T)}\Big)^{1/2}$$

for DG, SDG and SMDG, and the corresponding approximation property of the multiscale trial spaces (Proposition 3.4, Corollary 3.7, Theorems 4.3 and 4.6). All constants are independent of the mesh, the domain, κ and a.

The Lean files.

- `Showcase.lean` imports only Mathlib. It defines the setting and states each numbered result of the paper as its own theorem, in the paper's notation and with the paper's quantifiers, with `sorry`.
- `Showcase_WithProofs.lean` proves the same theorems from the library `LeanCode`.
- `comparator.json` configures `lake comparator`. It checks that the proofs establish exactly the statements of `Showcase.lean` and use only the axioms `propext`, `Classical.choice` and `Quot.sound`.
- The library `LeanCode` contains about 18 000 lines of proofs, organized as in the chapters below. Chapter 10 and the file `Audit/paper_correspondence.md` compare the formal statements with the paper.

Scope. The setting is the paper's, specialized to: dimension $d \in \{2,3\}$; every polynomial degree $k \geq 1$; conforming shape-regular affine simplicial meshes of a bounded connected Lipschitz domain; and constant coefficients $\kappa > 0$, $a \in \mathbb{R}^d$ (including $a = 0$), with no positive lower bound on κ.

Chapter 2

Results from Mathlib

The formalization builds on Mathlib. The following definitions and results of Mathlib are its main ingredients; in the dependency graph they are grey, and their edges show where they enter the proofs.

Definition 2.1 (Affine simplices)**.** [Lean: `Affine.Simplex, Affine.Simplex.closedInterior, Affine.Simplex.faceOpposite`] Affine simplices in a Euclidean space, their closed interiors and their faces.

Definition 2.2 (Hausdorff measure)**.** [Lean: `MeasureTheory.Measure.eclideanHausdorffMeasure`] The k-dimensional Hausdorff measure on a Euclidean space, normalized to agree with Lebesgue measure on k-dimensional subspaces; it is the measure on the faces of the cells.

Definition 2.3 (Multivariate polynomials)**.** [Lean: `MvPolynomial, MvPolynomial.pderiv`] Polynomials in d variables with real coefficients, and their partial derivatives.

Theorem 2.4 (Fubini–Tonelli theorem)**.** [Lean: `MeasureTheory.integral_prod, MeasureTheory.lintegral_prod, MeasureTheory.integral_integral_swap`] *Integrals over a product of σ-finite measure spaces are iterated integrals, in either order.*

Theorem 2.5 (Fundamental theorem of calculus)**.** [Lean: `intervalIntegral.integral_eq_sub_of_hasDerivAt`] $\int_a^b f' = f(b) - f(a)$ *when f has an integrable derivative on $[a, b]$.*

Theorem 2.6 (Bump functions and convolution)**.** [Lean: `ContDiffBump, HasCompactSupport.contDiff_convolution_left`] *There are smooth bump functions with compact support, and the convolution of a locally integrable function with a smooth compactly supported kernel is smooth.*

Theorem 2.7 (Fundamental lemma of the calculus of variations)**.** [Lean: `IsOpen.ae_eq_zero_of_integral_contDiff_smul_eq_zero`] *A locally integrable function on an open set whose integrals against all smooth compactly supported functions vanish is zero almost everywhere.*

Theorem 2.8 (Linear maps of finite-dimensional spaces)**.** [Lean: `LinearMap.injective_iff_surjective`] *A linear endomorphism of a finite-dimensional vector space is injective if and only if it is surjective.*

Chapter 3

Setting and definitions

All definitions are in `Showcase.lean`, namespace `BHSShowcase`. They are repeated verbatim in `LeanCode/Definitions.lean`.

Definition 3.1 (Admissible meshes)**.** [Lean: `BHSShowcase.MeshData, BHSShowcase.AdmissibleMesh, BHSShowcase.cellSet, BHSShowcase.facet, BHSShowcase.faceThickness`] A mesh consists of finitely many affine d-simplices T, a list of faces with an owner facet and an optional neighbour facet, and outward unit normals. It is *admissible with shape constant* $\rho \geq 1$ if:

- the domain Ω is bounded, connected, Lipschitz, and equal to the interior of the union of the cells;
- the cells have disjoint interiors and meet in common subsimplices (conformity);
- each cell contains a ball of radius r with $\operatorname{diam} T \leq \rho r$;
- each facet belongs to exactly one face;
- exactly the faces without a neighbour lie on $\partial\Omega$.

The face thickness is $h_\perp = (|T^+| + |T^-|)/(2|e|)$ on interior faces and $|T|/|e|$ on boundary faces.

Definition 3.2 (Weak jets and Sobolev traces)**.** [Lean: `BHSShowcase.WeakJet, BHSShowcase.HasSobolevTrace, BHSShowcase.BrokenField, BHSShowcase.BrokenH2, BHSShowcase.RealizesGlobalJet`] A *jet* assigns a function J_α to each multi-index α.

- It is a *weak* $H^m(\Omega)$ *jet* if $J_\alpha \in L^2(\Omega)$ for $|\alpha| \leq m$ and $J_{\alpha+e_i}$ is the distributional derivative $\partial_i J_\alpha$ for $|\alpha| < m$.
- Traces on the facets of a cell are L^2 limits of the facet restrictions of smooth functions whose derivatives converge in L^2 of the cell (`HasSobolevTrace`).
- A *broken field* has one jet per cell and one trace jet per facet. It is *broken* H^2 if every cell jet is H^2 with these traces.

Definition 3.3 (DG and SDG forms and norms)**.** [Lean: `BHSShowcase.dgForm, BHSShowcase.dgLoad, BHSShowcase.sdgForm, BHSShowcase.sdgLoad, BHSShowcase.dgNorm, BHSShowcase.sdgNorm, BHSShowcase.augmentedNorm, BHSShowcase.tau, BHSShowcase.jump, BHSShowcase.upwind`] For $s \in \{-1, 0, 1\}$ (symmetric, neutral, skew), the DG form is

$$B^{\mathrm{DG}}(\nu,\mu) = \sum_T \int_T (\kappa \nabla\nu\cdot\nabla\mu - \nu\, a\cdot\nabla\mu) + \sum_{e\in\mathcal{E}_h^o}\int_e \big(\llbracket\mu\rrbracket\cdot(a\nu^- - \kappa\nabla\nu^-) + s\kappa\nabla\mu^-\cdot\llbracket\nu\rrbracket\big)$$
$$+ \sum_{e\subset\partial\Omega}\int_e \big(s\kappa\nabla\mu\cdot n\,\nu - \kappa\nabla\nu\cdot n\,\mu + (a\cdot n)^+\mu\nu\big) + \varepsilon\sum_e\int_e \frac{\kappa}{h_\perp}\llbracket\mu\rrbracket\cdot\llbracket\nu\rrbracket,$$

with upwind values ν^- and the matching load L^{DG}. The SDG form adds $\sum_T \tau_T \int_T (-\kappa\Delta\nu + a\cdot\nabla\nu)(a\cdot\nabla\mu)$, with $\tau_T = \tau\min(h_T/|a|, h_T^2/\kappa)$. The norms are

$$\|\nu\|_{\mathrm{DG}}^2 = \sum_T \kappa\big(|\nu|_{H^1(T)}^2 + h_T^2|\nu|_{H^2(T)}^2\big) + \varepsilon\sum_e \frac{\kappa}{h_\perp}\|[\![\nu]\!]\|_{L^2(e)}^2 + \sum_e \||a\cdot n|^{1/2}[\![\nu]\!]\|_{L^2(e)}^2,$$

$$\|\nu\|_{\mathrm{SDG}}^2 = \|\nu\|_{\mathrm{DG}}^2 + \sum_T \tau_T\|a\cdot\nabla\nu\|_{L^2(T)}^2,$$

$$|\!|\!|\nu|\!|\!|_{\mathrm{SDG}}^2 = \|\nu\|_{\mathrm{SDG}}^2 + \sum_{e\in\mathcal{E}_h^o} \||a\cdot n|^{1/2}\nu^-\|_{L^2(e)}^2 + \sum_T \tau_T^{-1}\|\nu\|_{L^2(T)}^2.$$

Definition 3.4 (Local problems and transfer spaces)**.** [Lean: `BHSShowcase.localForm, BHSShowcase.localLoad, BHSShowcase.localEnergySq, BHSShowcase.CoarseField, BHSShowcase.localRHS, BHSShowcase.transferSpace`] On a cell T, with perturbation $\delta \ge 0$,

$$b_T(\nu,\mu) = \int_T (\kappa\nabla\nu\cdot\nabla\mu - \nu\, a\cdot\nabla\mu) + \sum_j \int_{F_j} \Big(-\kappa\nabla\nu\cdot n\,\mu + s\kappa\nabla\mu\cdot n\,\nu + \big(\varepsilon\tfrac{\kappa}{h_\perp} + (1+\delta)(a\cdot n)^+\big)\nu\mu\Big).$$

Let $\ell_T(\bar\nu, f; \mu)$ be the load with boundary data $\bar\nu$. The local energy $\|\nu\|_{\mathrm{SDG}(T)}^2$ is defined as in (4.7) of the paper. For a continuous piecewise-polynomial coarse field $\bar\nu \in \overline{V}_h$, the transfer $\mathfrak{T}_h(\bar\nu, f)$ is the solution of $b_T(\nu,\mu) = \ell_T(\bar\nu, f; \mu)$ on all cells. The trial space is $\mathfrak{T}_h(\overline{V}_h, f)$.

Definition 3.5 (Discrete problems, exact solutions, error indicator)**.** [Lean: `BHSShowcase.Vh, BHSShowcase.Vhbar, BHSShowcase.VhT, BHSShowcase.ExactSolution, BHSShowcase.IsExactSolution`] The DG, SDG and SMDG solutions are defined by the Galerkin equations (3.2), (3.13) and (4.4). An exact solution is a weak $H^{k+1}(\Omega)$ jet with L^2 data, realized by a broken field with traces, satisfying the equation a.e. in each cell and the boundary condition a.e. on $\partial\Omega$. The error indicator is $\big(\sum_T (|a|h_T^{2k+1} + \kappa h_T^{2k})|\varphi|_{H^{k+1}(T)}^2\big)^{1/2}$.

Chapter 4

The main theorems

Section 8 of `Showcase.lean` states each numbered result of the paper as a definition named after it (`Proposition_3_2`, ...), with the paper's quantifiers in the paper's order and in the paper's notation; Section 9 asserts them for $d \in \{2, 3\}$, $k \geq 1$ and $\rho \geq 1$. Constants are chosen before the mesh and the coefficients, which expresses that they are independent of h, κ and a.

Theorem 4.1 (Section 3 of the paper). [Lean: `BHS.proposition_3_2, BHS.proposition_3_2_skew, BHS.proposition_3_3, BHS.proposition_3_4, BHS.theorem_3_5, BHS.corollary_3_7`] *Propositions 3.2 (including its clause for $s = 1$), 3.3 and 3.4, Theorem 3.5 and Corollary 3.7, as stated in the paper. Theorem 3.5 and Corollary 3.7 hold for every streamline parameter $\tau > 0$; the inf-sup condition (3.24) is written as: for every $\nu \neq 0$ in V_h there is $\mu \neq 0$ in V_h with $B^{\mathrm{DG}}(\nu, \mu)/(\|\nu\|_{\mathrm{SDG}}\|\mu\|_{\mathrm{SDG}}) \geq \beta_{\mathrm{DG}}$.*

Proof. Each result is the corresponding stability or error estimate of the later chapters, translated from piecewise polynomials to the broken fields of V_h. The inf-sup quotient follows from the normalized witness $\|\mu\|_{\mathrm{SDG}} \leq \|\nu\|_{\mathrm{SDG}}$, $B^{\mathrm{DG}}(\nu, \mu) \geq \beta\|\nu\|^2_{\mathrm{SDG}}$, since the norm is definite on V_h. The DG error estimate needs only DG continuity and the DG inf-sup condition, so it holds for every $\tau > 0$. □

Theorem 4.2 (Section 4 of the paper). [Lean: `BHS.theorem_4_3, BHS.lemma_4_4, BHS.lemma_4_5, BHS.theorem_4_6`] *Theorem 4.3, Lemmas 4.4 and 4.5, and Theorem 4.6, as stated in the paper. Theorem 4.3 holds for every $\tau > 0$; Lemma 4.5 is stated on $H^1(T)$ without the H^2 term of $\|\cdot\|_{\mathrm{SDG}(T)}$.*

Proof. As for Section 3. Theorem 4.3 uses only the local inf-sup condition, the local Poincaré inequality and the interpolation estimate, so no smallness of τ is needed. □

Theorem 4.3 (Consistency, well-posedness and traces). [Lean: `BHS.consistency_DG_SDG, BHS.wellPosed_DG, BHS.wellPosed_SDG, BHS.wellPosed_local, BHS.wellPosed_SMDG, BHS.traceTheorem`] *The exact solution satisfies the DG and SDG equations; the problems (3.2), (3.13), (4.1) and (4.4) have unique solutions; every weak H^m jet on a cell has unique traces, and every weak $H^{k+1}(\Omega)$ jet is realized by a broken H^2 field. The last fact shows that the error estimates apply to every solution $\varphi \in H^{k+1}(\Omega)$.*

Proof. The local problems do not involve τ; their unique solvability is taken at $\tau = 1$, where the local inf-sup condition is available. □

Theorem 4.4 (Skew-symmetric penalty, Prop. 3.2). [Lean: `BHS.skewPenaltyCoercivity`] *For $s = 1$ and every $\varepsilon > 0$ there is $c > 0$ with $B^{\mathrm{DG}}(\mu, \mu) \geq c\|\mu\|^2_{\mathrm{DG}}$ on V_h, uniformly in the admissible mesh, $\kappa > 0$ and a.*

Proof. For $s = 1$ the consistency and adjoint terms cancel in $B^{\mathrm{DG}}_D(\mu, \mu)$. The convective part gives $\frac{1}{2}$ of the jump energy by integration by parts on each cell. □

Chapter 5

Geometry of simplicial meshes

Lemma 5.1 (Heights and shape regularity)**.** [Lean: `BHS.two_mul_le_height, BHS.cellSize_le_height, BHS.cellVolume_eq_height_mul_facetArea`] *If a cell contains a ball of radius r then every height $H_j \geq 2r$; hence $h_T \leq \frac{\rho}{2} H_j$. Moreover $|T| = \frac{1}{d} H_j |F_j|$.*

Proof. The inscribed ball lies between the facet hyperplane and the parallel hyperplane through the opposite vertex. The volume formula is the cone formula. □

Lemma 5.2 (Normals and face combinatorics)**.** [Lean: `BHS.normals_eq_outwardNormal, BHS.sum_cellFacets_eq_sum_faces`] *The mesh normals are the outward normals of the simplices. Sums over the facets of all cells can be regrouped into sums over faces (one or two sides each).*

Proof. Admissibility gives a bijection between cell facets and face sides. □

Lemma 5.3 (Interior points of faces)**.** [Lean: `BHS.exists_halfBall_subset, BHS.mem_interior_union_of_relint`] *Near a relative interior point of a facet, the cell contains a half ball. A point in the relative interior of an interior face is an interior point of the union of its two cells.*

Proof. Barycentric coordinates and the opposite normals of the two sides. □

Chapter 6

Analysis on simplices

Lemma 6.1 (Slicing and the cone formula)**.** [Lean: `BHS.integral_eq_mul_integral_slices, BHS.integral_closedInterior_cone, BHS.integral_radial_divergence`] *Integrals over a simplex can be computed along slices parallel to a facet, and as a cone over a facet with apex at the opposite vertex. Applied to the radial field, this gives the divergence identity for the radial vector field.*

Proof. Fubini along a direction, and the change of variables of a dilation. □

Theorem 6.2 (Divergence theorem on simplices)**.** [Lean: `BHS.integral_fderiv_eq_sum_facets, BHS.integral_inner_gradient_cell`] *For C^1 fields F on a neighbourhood of a simplex, $\int_T \operatorname{div} F = \sum_j \int_{F_j} F \cdot n_j \, \mathrm{d}\mathcal{H}^{d-1}$. In particular, Green's formula holds for smooth functions on the cells.*

Proof. Reduce to a single coordinate direction and integrate along slices; the facet integrals are identified with the $(d-1)$-dimensional Hausdorff measure. □

Lemma 6.3 (Polynomial inverse and trace inequalities)**.** [Lean: `BHS.exists_inverse_ball_sandwich, BHS.exists_cell_inverse, BHS.exists_cell_inverse_H2, BHS.exists_cell_trace, BHS.exists_cell_gradient_trace`] *For polynomials p of degree at most k on cells of ρ-regular meshes,*

$$h_T^2 \|\partial_i p\|_{L^2(T)}^2 \le C \|p\|_{L^2(T)}^2, \qquad h_T^2 |p|_{H^2(T)}^2 \le C |p|_{H^1(T)}^2, \qquad h_T \|p\|_{L^2(F_j)}^2 \le C \|p\|_{L^2(T)}^2,$$

with $C = C(d, k, \rho)$.

Proof. Equivalence of norms on the finite-dimensional space of polynomials on the unit ball. Then scale between the inscribed and the circumscribed ball. □

Lemma 6.4 (Trace inequality for smooth and Sobolev functions)**.** [Lean: `BHS.smooth_trace_ineq, BHS.hasSobolevTrace_trace_ineq`] $h_T \|v\|_{L^2(F_j)}^2 \le \frac{\rho}{2}\big((d+1)\|v\|_{L^2(T)}^2 + h_T^2 \|\nabla v\|_{L^2(T)}^2\big)$ *for C^1 functions, and hence for Sobolev traces.*

Proof. Apply the divergence theorem to the radial field $v^2(x - P_j)$. □

Theorem 6.5 (Density of smooth functions)**.** [Lean: `BHS.exists_smooth_approx`] *On an open convex set containing a ball, every weak H^m jet is the limit of smooth functions in all derivatives of order at most m.*

Proof. Dilate towards the centre of the ball, so that the dilated function is defined on a neighbourhood of the closure. Then mollify; weak derivatives commute with both operations. □

Theorem 6.6 (Traces exist, are unique, and agree across faces)**.** [Lean: `BHS.exists_hasSobolevTrace, BHS.hasSobolevTrace_unique, BHS.localTraceTheory, BHS.traceRealizationExists, BHS.traceAgreement`] *Every weak H^m jet on a cell ($m \ge 1$) has traces of order $< m$ on its facets, unique almost everywhere. Every weak $H^{k+1}(\Omega)$ jet is realized by a broken H^2 field. For a global weak jet, the traces of the two cells sharing an interior face agree almost everywhere.*

Proof. Existence and uniqueness: the trace inequality makes the facet restrictions of the smooth approximants Cauchy in L^2. Agreement: Green's formula on both cells and the weak derivative on Ω give $\int_e (t_T - t_S)\psi = 0$ for smooth ψ supported in $\operatorname{int}(T \cup S)$ away from the other facets. The fundamental lemma for the face measure gives $t_T = t_S$ a.e. there, and the rest of e is $\mathcal{H}^{d-1}$-null. □

Lemma 6.7 (Green's formula for Sobolev jets)**.** [Lean: `BHS.hasSobolevTrace_green`] *Green's formula holds on each cell for jets with Sobolev traces.*

Proof. Pass to the limit in the smooth formula. □

Lemma 6.8 (Poincaré–Wirtinger and polynomial moments)**.** [Lean: `BHS.poincare_convex`, `BHS.poincare_weakJet`, `BHS.exists_poly_moments`] *On a compact convex set K of diameter h, $\int_K |u - \bar{u}|^2 \le C h^2 \int_K |\nabla u|^2$, with C depending only on d and $h^d/|K|$. This holds for C^1 functions and for weak H^1 jets. For given moments m_α, $|\alpha| \le k$, there is a polynomial q of degree at most k with $\int \partial^\alpha q = m_\alpha$.*

Proof. Write $u(x) - u(y)$ as an integral along the segment and change variables; moments are fixed from the highest order downwards. □

Theorem 6.9 (Bramble–Hilbert)**.** [Lean: `BHS.exists_brambleHilbert`] *There is $C = C(d,k)$ such that for every cell of an admissible mesh and every weak H^{k+1} jet J on it, some polynomial q of degree at most k satisfies $h_T^{2j} |J - q|^2_{H^j(T)} \le C\, h_T^{2(k+1)} |J|^2_{H^{k+1}(T)}$ for all $j \le k+1$.*

Proof. Choose q so that all cell averages of $\partial^\alpha (J - q)$, $|\alpha| \le k$, vanish. Then iterate the weak Poincaré inequality. □

Theorem 6.10 (Scaled sup bound, $d \le 3$)**.** [Lean: `BHS.exists_sobolev_sup`, `BHS.sup_bound_convex`] *For $d \le 3$ and smooth u, at every point x of a cell, $|T|\, u(x)^2 \le C\big(\|u\|^2_{L^2(T)} + h_T^2 |u|^2_{H^1(T)} + h_T^4 |u|^2_{H^2(T)}\big)$ with $C = 3(4d+1)$.*

Proof. Average Taylor's formula with integral remainder over the cell. The remainder is controlled after the substitution $s = \sigma^2$, which needs $d \le 3$. □

Chapter 7

Interpolation

Lemma 7.1 (Lagrange interpolation on a simplex)**.** [Lean: `BHS.lagInterp, BHS.evalPoly_lagPoly_latNode, BHS.eq_zero_of_latNode, BHS.lagInterp_evalPoly, BHS.abs_evalPoly_lagPoly_le`] *Let the principal lattice of degree k have nodes x_λ, and let the Lagrange basis be*

$$\varphi_\lambda = \prod_i \prod_{j<\lambda_i} \frac{k b_i - j}{\lambda_i - j},$$

where b_i are the barycentric coordinates. Then $\varphi_\lambda(x_\mu) = \delta_{\lambda\mu}$ and the interpolant reproduces polynomials of degree at most k. Moreover $|\varphi_\lambda| \le k^k$ on the cell.

Proof. The nodal evaluation map is onto. Both spaces have dimension $\binom{d+k}{k}$, so it is also one-to-one. □

Lemma 7.2 (Face conformity)**.** [Lean: `BHS.exists_facetPerm, BHS.evalPoly_lagInterp_facet_eq, BHS.coarseField_lagInterp`] *If two cells share a facet, the interpolants of two functions that agree on the facet coincide on it. Hence cellwise interpolants of compatible functions form a coarse field in $\overline{V}_h$.*

Proof. The vertices of a shared facet are extreme points of both convex hulls, so the facet determines a permutation of the vertex indices. Under this permutation the barycentric coordinates agree on the facet. Basis functions with $\lambda_j \ge 1$ vanish on facet j; the remaining lattice nodes lie on the facet. □

Theorem 7.3 (Continuous representatives)**.** [Lean: `BHS.exists_continuousRep, BHS.facetMeasure_ball_pos, BHS.continuousRep_glue`] *Let $d \le 3$, and let J be a jet with H^2 traces on a cell. The smooth approximants of J converge uniformly on the closed cell. The limit w is continuous, agrees a.e. with J and with its facet traces, and satisfies the sup bound against every smooth comparison function. For a global jet, the representatives of neighbouring cells agree everywhere on the common face.*

Proof. The sup bound makes the approximants uniformly Cauchy. On a face, the two representatives agree a.e. by trace agreement. The face measure charges every relatively open subset of the face (homothety argument), so a.e. equality of continuous functions is equality. □

Theorem 7.4 (Interpolation estimate)**.** [Lean: `BHS.cell_interp_estimate, BHS.cell_localEnergy_le, BHS.exists_interpolationPremise, BHS.exists_interpolation`] *For $d \le 3$ there is C_I with the following property. For every exact solution there is a coarse field $c \in \overline{V}_h$ with $\sum_T \|\varphi - c\|^2_{\mathrm{SDG}(T)} \le C_I^2 \sum_T (|a| h_T^{2k+1} + \kappa h_T^{2k}) |\varphi|^2_{H^{k+1}(T)}$.*

Proof. Take $c|_T = I_T w_T$, the Lagrange interpolant of the continuous representative. With a Bramble–Hilbert polynomial q we have $I_T w - q = I_T(w - q)$. Its nodal values are bounded by the sup bound, so the inverse inequalities give the cell terms. The facet terms follow from the sup bound and $|F_j|/|T| \le d\rho/(2h_T)$. □

Chapter 8

Stability of the discrete methods

Proposition 8.1 (DG coercivity, Prop. 3.2)**.** [Lean: `BHS.exists_dgCoercivity`] *There is ε_0 such that for $\varepsilon \geq \varepsilon_0$, $B^{\mathrm{DG}}(\mu,\mu) \geq c\|\mu\|^2_{\mathrm{DG}}$ on V_h, with c independent of the mesh, κ, a and ε.*

Proof. For polynomial fields, $B^{\mathrm{DG}}(p,p) = \kappa|p|^2_{H^1} + \frac{1}{2}\|p\|^2_C + \varepsilon J_0(p) - (1-s)\kappa\sum_e \int_e \nabla p^- \cdot [\![p]\!]$, with $J_0(p) = \sum_e \frac{\kappa}{h_\perp}\|[\![p]\!]\|^2_{L^2(e)}$. Young's inequality, the gradient trace inequality and $h_\perp \lesssim h_T$ bound the last term by $\frac{1}{2}\kappa|p|^2_{H^1} + CJ_0(p)$, which is absorbed for $\varepsilon \geq \varepsilon_0$. The constant $1/(2(1+K))$ is uniform in $\varepsilon \geq \varepsilon_0$. □

Proposition 8.2 (SDG coercivity, Prop. 3.3)**.** [Lean: `BHS.exists_sdgCoercivity`] *There are $\varepsilon_0, \tau_{\max} > 0$ such that for $\varepsilon \geq \varepsilon_0$ and $0 \leq \tau \leq \tau_{\max}$, $B^{\mathrm{SDG}}(\mu,\mu) \geq c\|\mu\|^2_{\mathrm{SDG}}$ on V_h.*

Proof. $\tau_T\|\kappa\Delta\mu\|^2 \lesssim \tau\|\mu\|^2_D$ by the H^2 inverse inequality; absorb the SUPG cross term for small τ. □

Proposition 8.3 (Continuity, Prop. 3.3)**.** [Lean: `BHS.exists_continuity`] *For all broken H^2 fields U and V, $|B^{\mathrm{DG}}(U,V)| \leq C|\!|\!|U|\!|\!|_{\mathrm{SDG}}\|V\|_{\mathrm{SDG}}$ and $|B^{\mathrm{SDG}}(U,V)| \leq C|\!|\!|U|\!|\!|_{\mathrm{SDG}}\|V\|_{\mathrm{SDG}}$, with C independent of the mesh, κ and a.*

Proof. Bound the diffusive, convective and SUPG parts separately (paper (3.19)–(3.21)), using the trace inequality for the face gradients. □

Theorem 8.4 (DG inf-sup in the SDG norm, Thm. 3.5)**.** [Lean: `BHS.exists_dgInfSup`] *There is ε_0 such that for all $\varepsilon \geq \varepsilon_0$ and $\tau > 0$ there is $\beta > 0$ with: for every $\nu \in V_h$ some $\mu \in V_h$ satisfies $\|\mu\|_{\mathrm{SDG}} \leq \|\nu\|_{\mathrm{SDG}}$ and $B^{\mathrm{DG}}(\nu,\mu) \geq \beta\|\nu\|^2_{\mathrm{SDG}}$.*

Proof. Take $\mu = \nu + \gamma\sum_T \tau_T(a\cdot\nabla\nu)|_T$ as in the paper. Bound the extra terms by inverse and trace inequalities, then normalize. □

Lemma 8.5 (Local inf-sup, Lemma 4.4)**.** [Lean: `BHS.exists_localInfSup`] *There is ε_0 such that for $\varepsilon \geq \varepsilon_0$, $\tau > 0$ and every bound $\delta_{\max} \geq 0$ there is $\beta_L > 0$ such that the local forms b_T satisfy the normalized inf-sup condition in $\|\cdot\|_{\mathrm{SDG}(T)}$ for all $\delta \in [0, \delta_{\max}]$.*

Proof. As in Theorem 3.5, the witness is $\nu + \gamma\tau_T(a\cdot\nabla\nu)$, normalized. It uses local coercivity of b_T for $\varepsilon \geq \varepsilon_0$, a lower bound in the streamline direction, and the inherited face thickness, which is comparable to h_T by shape regularity. □

Lemma 8.6 (Local Poincaré inequality, Lemma 4.5)**.** [Lean: `BHS.exists_localPoincare`] *$\tau_T^{-1}\|\nu\|^2_{L^2(T)} \leq C\|\nu\|^2_{\mathrm{SDG}(T)}$ (without the Hessian term) for every H^1 jet ν with traces.*

Proof. Instead of the paper's Lipschitz function η, apply the divergence theorem to $(a\cdot(x-v_0))\,w^2$, where v_0 is a vertex; this gives $|a|^2\|w\|^2_T \leq 2|a|h\sum_j \||a\cdot n_j|^{1/2}w\|^2_{F_j} + 4h^2\|a\cdot\nabla w\|^2_T$. The radial identity gives the diffusive bound $d^2\|w\|^2_T \leq 2dh\|w\|^2_{F_j} + 4h^2|w|^2_{H^1}$. Both hold for C^1 functions and pass to Sobolev jets through their smooth approximants. □

Theorem 8.7 (Consistency). [Lean: `BHS.consistency_of`] *For every exact solution φ (with $k \geq 1$), $B^{\mathrm{DG}}(\varphi, \mu) = L^{\mathrm{DG}}(\mu)$ and $B^{\mathrm{SDG}}(\varphi, \mu) = L^{\mathrm{SDG}}(\mu)$ for all $\mu \in V_h$.*

Proof. Green's formula on each cell. The interior face terms cancel by trace agreement; the boundary terms use the boundary condition. □

Theorem 8.8 (Unique solvability). [Lean: `BHS.existsUnique_dgSolution`, `BHS.existsUnique_sdgSolution`, `BHS.existsUnique_smdgSolution`, `BHS.existsUnique_transfer`] *DG, SDG, SMDG and the local transfer problems have unique solutions.*

Proof. The discrete norms are definite on V_h. A field with vanishing energy is piecewise constant with vanishing jumps (including the boundary). The divergence theorem for $x \mapsto \langle e_0, x \rangle$ on every cell then gives $\sum_T c_T^2 |T| = 0$. Coercivity and the inf-sup conditions give injectivity, and finite dimension gives surjectivity. SMDG is posed on the linear space $\mathfrak{T}_h(\overline{V}_h, 0)$, shifted by a particular transfer. □

Chapter 9

Error estimates

Lemma 9.1 (From local energies to the global norm)**.** [Lean: `BHS.ErrorEst.augmentedNorm_sq_le_sum_localEnergySq`] *For every broken H^2 field z,*

$$\|\!|z|\!\|_{\mathrm{SDG}}^2 \le (3 + 2\rho d + C_P) \sum_T \|z\|_{\mathrm{SDG}(T)}^2.$$

Proof. Jump energies are bounded by the facet traces of both sides; upwind traces are charged once; the τ_T^{-1} term is bounded by Lemma 4.5. □

Theorem 9.2 (Error estimates: Prop. 3.4, Cor. 3.7, Thms. 4.3 and 4.6)**.** [Lean: `BHS.exists_errorEstimates, BHS.exists_transferApproximation`] *Assume SDG stability, the DG and local inf-sup conditions, the local Poincaré inequality, consistency, and an interpolation premise with constant C_I. Then the SDG, DG and SMDG errors in $\|\cdot\|_{\mathrm{SDG}}$, and the transfer approximation error in $\|\!|\cdot|\!\|_{\mathrm{SDG}}$, are bounded by C times the error indicator. Here C depends only on d, ρ, ε and the given constants.*

Proof. Galerkin orthogonality, together with coercivity (SDG, SMDG) or the inf-sup witness (DG), and continuity. For the transfer, consistency of the local problems gives an exact identity for $b_T(c - \nu, \mu)$; the local inf-sup condition and Lemma 9.1 conclude. □

Chapter 10

Correspondence with the paper

The detailed comparison is in `Audit/paper_correspondence.md`. In summary:

- All numbered results of Sections 3 and 4 are stated: Propositions 3.2–3.4, Theorem 3.5, Corollary 3.7, Theorem 4.3, Lemmas 4.4 and 4.5, and Theorem 4.6. They have the paper's uniformity: constants independent of the mesh, the domain, κ and a. The quantifier order is the paper's.
- Scope: simplicial meshes, constant coefficients, and $\kappa > 0$.
- Repairs of printed formulas: the κ-weight of the jump term in (3.7) is linear; the sign of the s-term follows (3.2); continuity (3.17) holds on broken H^2; $h_\perp$ in (4.2) is the global face thickness; Lemma 4.5 is stated on H^1 without the Hessian term.
- Not formalized: Section 5 (numerics), Remark 3.6, and the open problem of Remark 4.7.